

Math 421

Wednesday, October 8

Content info &
study materials
posted to
Canvas.

Announcements

- Homework 5 due Friday
- Grace's Drop-In Hours
 - **Today:** 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Exam 1 – 10/15 5:45-7:15 pm
 - If you have a conflict, email me (if you haven't already) and **fill out makeup exam request form** (linked from Exam Information page on Canvas) ASAP!
 - If you are a McBurney student, sign up to take the exam at the testing center through RegisterBlast.

Algebraic Limit Laws

Theorems 9.2-9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then,

i. $\lim ks_n = ks$ for all $k \in \mathbb{R}$.

ii. $\lim s_n + t_n = s + t$.

iii. $\lim s_n t_n = st$.

iv. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$

v. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$

Want $|s_n| \geq m$

FALSE Proof of iv. if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{1}{s_n} = \frac{1}{s}$.

Activity: Critique the proof below.

Proof. Fix $\epsilon > 0$. Since (s_n) is convergent, there is a constant $M > 0$ such that $|s_n| \leq M$ for all n . Since $\lim s_n = s$, there exists $N > 0$ such that if $n > N$, then $|s_n - s| < \epsilon M |s|$. Assume $n > N$. Then

$$\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s - s_n}{s_n s} \right| \leq \frac{|s_n - s|}{M |s|} < \frac{\epsilon M |s|}{M |s|} = \epsilon$$

Since $\epsilon > 0$ was arbitrary, $\lim \frac{1}{s_n} = \frac{1}{s}$.

$\rightarrow \frac{1}{|s_n|} \geq \frac{1}{M}$

Proof of iv.

Lemma 9.5: Let $\lim s_n = s$, if $s_n \neq 0$, for all n , and if $s \neq 0$, then

$$\lim \frac{1}{s_n} = \frac{1}{s}.$$

Discussion.

- Since $\lim s_n = s$, there exists N such that $n > N$ $|s_n - s| < \frac{|s|}{2}$

$$|s| - |s_n| \leq |s - s_n| = |s_n - s| < \frac{|s|}{2} \Rightarrow |s| - |s_n| < \frac{|s|}{2} \Rightarrow$$

$$\frac{|s|}{2} - |s_n| < 0 \Rightarrow |s_n| > \frac{|s|}{2}.$$

- Want $|\frac{1}{s_n} - \frac{1}{s}| < \epsilon.$

$$\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s - s_n}{s_n s} \right| < \frac{|s - s_n|}{\frac{|s|}{2} \cdot |s|} < \epsilon$$

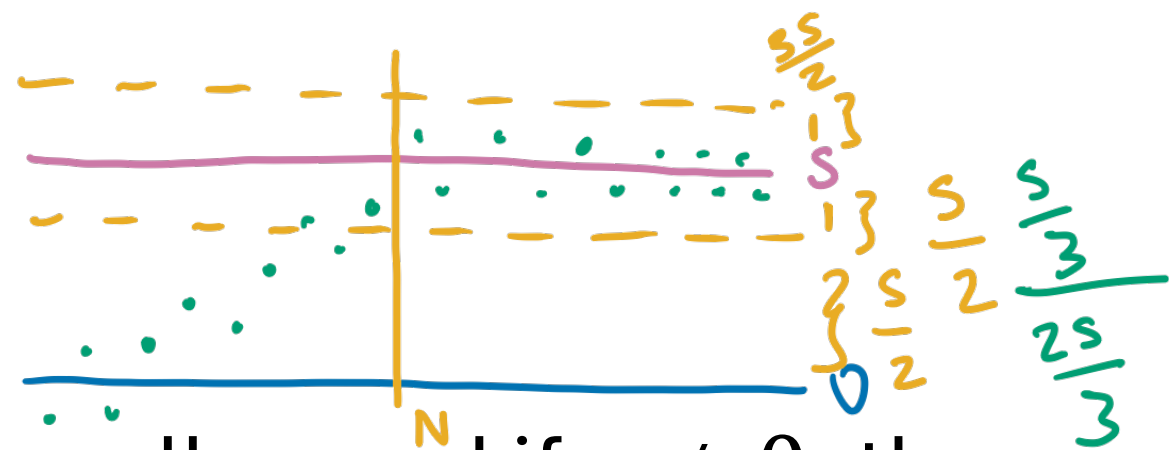
$$\Leftrightarrow |s - s_n| = |s_n - s| < \frac{\epsilon |s|^2}{2}$$

$$\frac{2|s - s_n|}{|s|^2} < \epsilon$$

Goal: $|s_n| > m > 0$

$$n > N, \quad s_n > \frac{s}{2}$$

$$s_n > \frac{2s}{3}$$



$$|s_n - s| < \min \left\{ \frac{|s|}{2}, \frac{\epsilon |s|^2}{2} \right\} \text{ want } < |s| \text{ would be } \frac{|s|}{3} \text{ would not be } |s| - 1$$

Proof of iv.

Lemma 9.5: Let $\lim s_n = s$, if $s_n \neq 0$, for all n , and if $s \neq 0$, then

$$\lim \frac{1}{s_n} = \frac{1}{s}.$$

Proof. Let $\varepsilon > 0$. Since $\lim s_n = s$, there exists N such that $n > N$ implies $|s_n - s| < \min\left\{\frac{|s|}{2}, \frac{\varepsilon |s|^2}{2}\right\}$. Assume $n > N$,

$$\text{then } |s_n - s| < \frac{|s|}{2} \Rightarrow |s| - |s_n| \leq |s - s_n| = |s_n - s| < \frac{|s|}{2}$$

$$\Rightarrow |s_n| > \frac{|s|}{2} \quad \text{so}$$

$$\left| \frac{1}{s_n} - \frac{1}{s} \right| = \left| \frac{s - s_n}{s_n s} \right| < \frac{|s - s_n|}{\frac{|s|}{2} \cdot |s|} < \frac{\varepsilon \frac{|s|^2}{2}}{\frac{|s|^2}{2}} = \varepsilon.$$

since $\varepsilon > 0$ was arbitrary $\lim \frac{1}{s_n} = \frac{1}{s}$.

Proof of v.

Theorem 9.6: Let $\lim s_n = s$ and $\lim t_n = t$. Then, if $s_n \neq 0$, for all n , and if $s \neq 0$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

Proof.

$$\lim \frac{t_n}{s_n} = \lim \frac{1}{s_n} \cdot t_n = \frac{1}{s} \cdot t = \frac{t}{s}.$$

Example: $\lim \frac{4n^3+3n}{n^3-6} = 4$

Chapter 8
example 3.

Past: using precise limit definition

$$\left| \frac{4n^3+3n}{n^3-6} - 4 \right| \dots N > \underline{\hspace{2cm}}$$

Now:

$$\frac{4n^3+3n}{n^3-6} = \frac{\cancel{n^3} (4 + \frac{3}{n^2})}{\cancel{n^3} (1 - \frac{6}{n^3})}$$

$$\lim 4 + \frac{3}{n^2} = \lim 4 + 3 \lim \left(\frac{1}{n}\right)^2 = 4 + 3 \cdot (0)$$

$$\lim 1 - \frac{6}{n^3} = \lim 1 - 6 \lim \left(\frac{1}{n}\right)^3 = 1 - 6(0)$$

$$\Rightarrow \lim \frac{4n^3+3n}{n^3-6} = \frac{4}{1}$$